



Birla Vishvakarma Mahavidyalaya Engineering College

The Robotics Society, India Student chapter

Report on

Event Report: “ROS Workshop and Hands-on Implementation”

DATE AND DAY:- 3rd & 10th September, 2025,

WEDNESDAY

A two-day technical event on **Robot Operating System (ROS)** was organized with the objective of providing students with both theoretical understanding and practical exposure to robotics software and hardware integration. The workshop mainly focused on the fundamentals of ROS, working of TurtleBot, implementation of SLAM (Simultaneous Localization and Mapping), and real-time robotic control using ROS.

The event was highly interactive and application-oriented, allowing participants to bridge the gap between theory and practical robotics implementation.

Day 1: Introduction to TurtleBot and SLAM Implementation

On the first day, the session began with an introduction to ROS (Robot Operating System) by **Prof.(Dr) D.M Patel**. Participants were introduced to concepts such as:

- ROS architecture
- Nodes and topics
- Publishers and subscribers
- Services and messages
- Sensor integration

The resource person then explained the working of TurtleBot, which is a popular open-source mobile robot platform widely used for learning and research in robotics.

TurtleBot Overview

The explanation covered:

- Hardware components (LiDAR sensor, camera, motors, controller)
- Communication between sensors and ROS nodes
- Real-time data publishing
- Mapping and navigation basics

Implementation of SLAM

A major highlight of Day 1 was the practical demonstration of SLAM (Simultaneous Localization and Mapping).

The team demonstrated how TurtleBot was used to perform mapping of the ground floor area of the building. The robot moved around the floor, collecting sensor data through LiDAR. This data was processed using SLAM algorithms in ROS to:

- Detect obstacles
- Create a 2D map of the environment
- Identify robot position in real-time

Participants observed how the live map was generated on the system screen while the robot was navigating. The mapping process clearly showed how robotics integrates sensors, algorithms, and software to solve real-world navigation problems.

This session helped students understand:

- Importance of localization
- Environmental perception
- Real-time mapping techniques
- Practical use of SLAM in autonomous systems



Day 2: Robotic Kit Implementation and Project Development

The second day was fully focused on hands-on learning and implementation.

Participants were divided into teams of three members each, encouraging teamwork, coordination, and problem-solving.

Each team was provided with a robotic kit, which included:

- Motors
- Motor drivers
- Sensors
- Controller board
- Chassis

Controlling the Robot Using ROS

Under guidance, students connected the robotic kit to ROS and learned how to:

- Write simple ROS nodes
- Control motor movement
- Publish velocity commands
- Subscribe to sensor data
- Implement basic motion control

The teams implemented robot control using ROS commands and successfully made the robot:

- Move forward and backward
- Turn left and right
- Stop using command inputs

This practical session strengthened understanding of:

- ROS communication system
- Hardware-software integration
- Motor control logic
- Team collaboration

Mini Project Development

After learning basic control, each team was instructed to develop a small working project using their robotic kit.

The objective was to apply the learned concepts creatively. Teams worked on:

- Basic navigation logic
- Obstacle detection
- Controlled movement patterns

Students actively discussed, divided tasks among team members, debugged errors, and tested their implementations.

This activity enhanced:

- Technical confidence
- Logical thinking
- Team coordination
- Practical problem-solving ability

Learning Outcomes

By the end of the two-day event, participants gained:

- Clear understanding of ROS fundamentals
- Practical knowledge of TurtleBot operation
- Real-time experience with SLAM mapping
- Hands-on experience in robotic control
- Team-based project development skills

The workshop successfully combined theoretical concepts with real-world implementation, making it highly beneficial for students interested in robotics and automation.



Conclusion

The ROS event was informative, engaging, and practically enriching. It provided students with exposure to real-world robotics systems and modern automation tools. The combination of SLAM mapping using TurtleBot and hands-on robotic kit implementation made the event both educational and inspiring.

Overall, the workshop enhanced participants' understanding of robotic systems and motivated them to explore further in the field of robotics and intelligent automation.

Robot Operating System (ROS) Work Shop

September 3 & 10, 2025

Venue: D101, BVM, Vallabh Vidyanagar

Attendance Sheet

Team No.	Sr No	Name	ID	3/9/2025		10/9/2025	
				Session 1 2-4	Session 2 4-6	Session 1 2-4	Session 2 4-6
Team 1	1	Namita Patel		AB	AB		
	2	Arya Patel	24EL303	AB	AB	AB	✓
	3	Masoom Abhani	25CP993	AB	AB	AB	✓
	4	Smit Talaviya		AB	AB		
	5	Sahil Karena		AB	AB		
	6	Aditya Panchal		AB	AB		
	7	Moradiya Krisha B.		✓	✓		
Team 2	8	Pandit Yatri	24EL081	Yatri	Yatri	Yatri	Yatri
	9	Dharmik Suthar	25EL077	AB	AB	DS	✓
	10	Het Desai	23EC51K	H. Desai	H. Desai		
	11	Darshan Jagad	23EC407			Darshan	✓
	12	Deep Rupapara	24EL311	Deep R.	Deep R.	Deep R.	✓
	13	Shivam Machhi	24EL319	Shivam	Shivam	Shivam	✓
	14	Satheesh Nadar		AB	AB		
Team 3	15	Bhakti Gohel					
	16	Heet Vekanya	23EL057	Heet	Heet	Heet	✓
	17	Aaditya Popli		AB	AB		✓
	18	Chauhan Bhumi C.		AB	AB		
	19	Kishor Solanki	24EL315	Kishor	Kishor	Kishor	✓
	20	Bhagirathsinh Mori	23EL016		✓	✓	✓
	21	Ruchit N Soni		AB	AB		
Team 4	22	Prem Kansara		AB	AB		
	23	Aaditya Shinde		AB	AB		
	24	Himanshu Kumhar		AB	AB		
	25	Patil Ganesh Vasudev		AB	AB		
	26	Krish Ambani	24ME034	Krish	Krish	Krish	Krish
	27	Sevaramani Vidhi P.		AB	AB	Sevaramani	✓
	28	Rufansh P. Panchal		AB	AB	Rufansh	✓
Team 5	29	Yash Gandhi		✓	✓		
	30	Saxam Desai	25EL039	AB	AB	Saxam	✓
	31	Manmeet Joshi		✓	✓	✓	✓
	32	Manthan Rami		✓	✓	✓	✓
	33	Vaghamshi Jatin J.		✓	✓		
	34	Fahim Saiyed		AB	AB	Saiyed	✓
	35	Vedang Patel		✓	✓	Vedang	✓
Team 6	36	Mourya Ankit H.		✓	✓		
	37	Prachi Bachani		AB	AB		
	38	Yagnik Vadaliya		✓	✓		
	39	Maitulya Desai	25ME119	AB	AB	✓	✓
	40	Nikunj Suthar		AB	AB		
	41	Thakur Dhruv		AB	AB		
	42	Kishan Mori	23EC450	Kishan	Kishan	Kishan	Kishan
Team 7	43	Pambhar Hitanshi		AB	AB		
	44	Sahil Makwana					

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			Session 1 2-4	Session 2 4-6	Session 1 2-4	Session 2 4-6
45	Anvi Joshi					
46	Meet Mori		✓	✓		
47	Anshul Majmudar		AB			
48	Riya Natubhai Patel	24ELO92	Riya	Riya	Riya	Riya
49	Khushi kushwana	24ELO69	Khushi	Khushi	Khushi	Khushi
50	Parmar Dev R		AB			
51	Kavya Shah		AB		Kavya	✓
52	Dev Tank	24ME021	Dev	Dev	Dev	✓
53	Maitulya Desai		✓	✓		
54	Jash Patel	25EP047	Jash	Jash	Jash	Jash
55	Ansh Patel	25EC423	Ansh	Ansh	Ansh	✓
56	Kushal Gobariya	25EL432	Kushal	Kushal	Kushal	✓
57	Ganesh Patil		✓	✓		
58	Radhika Savaliya	25ELO52	Radhika	Radhika	Radhika	✓
59	Krishna Kansagra	25ELO47	Krishna	Krishna	Krishna	✓
60	Mathuriya Dev		✓	✓		
61	Shah Aaha		✓	✓		
62	Anvi Joshi		✓	✓		
63	Nihal Patil		✓	✓		

35/62
Attended
on 3/9/25
Uppala

64	Karthik Smith	25ME439			✓	✓
65	Shah Aesha	24ME018			Shah Aesha	✓
66	Tajpurni Munawar	25EC436			Tajpurni	✓
67	Renish Bhunani	24ELO48			Renish	✓
68	Biswal Divisha	24ELO40			Biswal	✓
69						
70						
71	Satsang Shete	24ME027			✓	✓
72	Jeel Gunjapati	24ME012			✓	✓
73	Pranav Rishabh	24ME030			✓	✓
74						
75						
76						
77						
78						
79						

37 Attended
on 10/9/25
Uppala